INCH-POUND
MIL-DTL-21427C(SH)
W/AMENDMENT 1
21 January 2020
SUPERSEDING
MIL-DTL-21427C(SH)
17 July 2018

DETAIL SPECIFICATION

STRAINER, ASSEMBLIES, MAIN STEAM, HIGH PRESSURE (SIZES 4 INCHES AND ABOVE)

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers high pressure steam strainer assemblies in sizes 4 inches and above intended primarily for use in main steam systems to protect power equipment and throttle valves from dirt, sediment, pipe scale, and foreign objects.

1.2 Classification. Strainer bodies covered by this specification are of the following compositions as specified (see 6.2):

a. Composition A - 2¼ percent chromium, 1 percent molybdenum alloy
b. Composition B - 1¼ percent chromium, ½ percent molybdenum alloy
c. Composition C - Carbon steel

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3. and 4. of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3. and 4. of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-1222	-	Studs, Bolts, Screws and Nuts for Applications where a High Degree of Reliability is Required; General Specification for
MIL-G-24716	-	Gaskets, Metallic-Flexible Graphite, Spiral Wound

(Copies of these documents are available online at https://quicksearch.dla.mil.)

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 4730

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2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

S9074-AR-GIB-010/278 - Requirements for Fabrication Welding and Inspection, and Casting Inspection and Repair for Machinery, Piping, and Pressure Vessels

T9074-AS-GIB-010/271 - Requirements for Nondestructive Testing Methods

(Copies of these documents are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. These documents are available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B16.1/ASME B16.5 - Pipe Flanges and Fittings Package

Boiler and Pressure Vessel Code, Section I, Rules for Construction of Power Boilers

Boiler and Pressure Vessel Code, Section II, Materials

Boiler and Pressure Vessel Code, Section VIII, Rules for the Care of Power Boilers

(Copies of these documents are available online at www.asme.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ANSI/ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at www.asq.org.)

ASTM INTERNATIONAL

ASTM A105/A105M - Standard Specification for Carbon Steel Forgings for Piping Applications

ASTM A106/A106M - Standard Specification for Seamless Carbon Steel Pipe for High-Temperature Service

ASTM A182/A182M - Standard Specification for Forged or Rolled Alloy and Stainless Steel Pipe Flanges, Forged Fittings, and Valves and Parts for High-Temperature Service

ASTM A193/A193M - Standard Specification for Alloy-Steel and Stainless Steel Bolting for High Temperature or High Pressure Service and Other Special Purpose Applications

ASTM A194/A194M - Standard Specification for Carbon Steel, Alloy Steel, and Stainless Steel Nuts for Bolts for High Pressure or High Temperature Service, or Both

ASTM A216/A216M - Standard Specification for Steel Castings, Carbon, Suitable for Fusion Welding, for High-Temperature Service

ASTM A217/A217M - Standard Specification for Steel Castings, Martensitic Stainless and Alloy, for Pressure-Containing Parts, Suitable for High-Temperature Service

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ASTM A240/A240M - Standard Specification for Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels and for General Applications

ASTM A335/A335M - Standard Specification for Seamless Ferritic Alloy-Steel Pipe for High-Temperature Service

(Copies of these documents are available online at www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. Materials shall be as specified in table I.

TABLE I. Materials.

Name of part	Material form	Applicable documents		
		Composition A	Composition B	Composition C
Body, covers	Forgings	ASTM A182/A182M, grade F-22	ASTM A182/A182M, grade F-11	ASTM A105/A105M, grade II
	Castings	ASTM A217/A217M, grade P-22	ASTM A217/A217M, grade WC-6	ASTM A216/A216M, grade WCB
Welded bosses drain connection	Pipe	ASTM A335/A335M, grade P-22	ASTM A335/A335M, grade P-11	ASTM A106/A106M
Basket, rims	Plate	ASTM A240/A240M	ASTM A240/A240M	ASTM A240/A240M
Gaskets	Spiral wound	MIL-G-24716	MIL-G-24716	MIL-G-24716
Bolts and washers	Alloy steel	MIL-DTL-1222, grade B16	MIL-DTL-1222, grade B16	ASTM A193/A193M, grade B7
Nuts	Alloy steel	MIL-DTL-1222, grade 7, heavy hex	MIL-DTL-1222, grade 7, heavy hex	ASTM A194/A194M, grade 4 or 7

3.1.1 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.2 Construction. The strainer assemblies shall be composed of a body with inlet connections, closure, strainer basket assembly, bosses for installation of thermometers, and either a single or dual outlet as specified (see 6.2). In no case shall the basket assembly be permanently attached to the end closure. Where strainer body inlet or outlet connections are welded, they shall be limited to ½ the size of the strainer body for services 600 pounds per square inch (lb/in^2) and above or 425 degrees Fahrenheit (°F) and above.

3.2.1 Pressure temperature ratings. The strainer bodies shall be in accordance with ASME Boiler and Pressure Vessel Code, sections I, II, and VIII, as applicable. The design pressure shall be equal to 1.05 times maximum system pressure and the design temperature shall be 10 °F plus maximum system temperature (see 6.2).

3.2.2 Temperature ratings. The maximum allowable temperatures for strainers specified herein shall be as follows (see 6.2):

a. Composition A - 1,050 °F

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b. Composition B - 1,000 °F

c. Composition C - 775 °F

3.2.3 Sizes. The sizes of strainer assemblies, number of inlets and outlets and their sizes, and whether straight, wye, angle, or manifold shape shall be as specified (see 6.2).

3.2.4 Strainer baskets. Strainer baskets shall be made of drilled plate. The welded seam shall not be drilled. The net open area through the basket shall be at least 2 times the inlet pipe area. The plate shall have 3/32-inch maximum drilled openings. The strainer basket shall be strong enough to prevent failure in case it is plugged a sufficient amount to produce a pressure drop across the basket equivalent to 25 percent of the working pressure. The wall thickness for the strainer basket shall be determined from the formula in accordance with section VIII of ASME Boiler and Pressure Vessel Code. In no case shall the wall thickness be less than ¼ inch. The ligaments between the perforations shall be determined from the formula in accordance with section I of ASME Boiler and Pressure Vessel Code. The ligament efficiency shall be not less than 30 percent and not more than 50 percent. The holes in the basket shall be arranged so that the maximum continuous unperforated area is in transverse or circumferential direction. The welded seam of the strainer basket shall not be located in the direct line of the main steam flow through the strainer.

3.2.5 Fabrication. Welding shall be in accordance with S9074-AR-GIB-010/278.

3.2.6 End connections. Unless otherwise specified (see 6.2), end connections shall be butt welding ends prepared in accordance with ASME B16.1/ASME B16.5. For flanged strainer assemblies, flanges in accordance with ASME B16.1/ASME B16.5 shall be specified (see 6.2). Flange faces, when flanges are specified, shall be of a finish with a circular lay (concentric or phonographic) having a roughness not exceeding 500 roughness height rating (RHR) produced by machining cuts of 0.003 inch maximum depth with not less than 40 cuts per inch of face width. For special installations involving radioactive service or hazardous fluids (toxic or explosive) where a finer finish is required, the requirements shall be the same except that the finish shall be 125 RHR maximum. For flanges where the flange face cannot be turned and tool marks run across the flange face, the surface finish shall have a maximum RHR of 125.

3.2.7 End closure. End closure shall be flanged.

3.2.8 Drain connections. Each strainer shall be provided with a drain connection. The drain connection shall be located so as to permit drainage of the strainer.

3.2.9 Steam flow. Unless otherwise specified (see 6.2), the strainer shall be constructed so that the steam flow through the strainer basket is from the inside of the basket to the outside of the basket.

3.2.10 Interchangeability. In no case shall parts be physically interchangeable or reversible unless such parts are also interchangeable or reversible with regard to function, performance, and strength.

3.3 Marking. Strainer bodies shall have the pressure rating and manufacturer's name or trademark cast or forged integral with the strainer body. No metal stamping shall be permitted on the finished strainer bodies or surfaces subjected to high stresses in service.

3.4 Nondestructive testing. Strainers shall be 100 percent radiographically inspected in accordance with S9074-AR-GIB-010/278.

4. VERIFICATION

4.1 Sampling for examination. A random sample of strainers shall be selected from each lot offered to the Government in accordance with ANSI/ASQ Z1.4 at inspection level II and the Acceptance Quality Limit (AQL) specified (see 6.2).

4.1.1 Visual examination. Sample strainers shall be visually examined in accordance with table II. Strainers not meeting the requirements of table II shall be rejected.

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TABLE II. Examination requirements.

Examination	Test paragraph	Requirement paragraph
Dimensions	4.1.2	3.2.3, 3.2.4
Materials	4.1.3	3.1
Design and construction	4.1.3	3.2, 3.3
Pressure and temperature	4.1.3	3.2.1, 3.2.2

4.1.2 Dimensional examination. Sample strainers shall be examined to determine conformance with the dimensional requirements specified herein (see 3.2.3 and 3.2.4).

4.1.3 Design. Sample strainers shall be examined to determine conformance with material (see 3.1), design and construction (see 3.2 and 3.3), and pressure and temperature (see 3.2.1 and 3.2.2) requirements specified herein.

4.2 Tests.

4.2.1 Hydrostatic tests. Each of the strainers shall be subjected to the tests specified in 4.2.1.1 for strength and porosity. The water temperature shall not exceed 100 °F.

4.2.1.1 Shell test. Strainer shall be subjected to pressures determined by the following formula for 10 minutes. Any weeping, leakage, or permanent deformation shall be cause for rejection:

$P_h = 1.5\ (P_d \times \frac{S_r}{S_o})$
Where:
P_h = Required hydrostatic pressure (lb/in^2 gauge)
P_d = Design pressure of the strainer (lb/in^2 gauge)
S_o = Allowable stress at design temperature (lb/in^2)
S_r = Allowable stress at room temperature (lb/in^2)

4.2.2 Radiographic inspection. Pressure-containing castings and flanges of strainers shall be 100 percent radiographically inspected. Inspection shall be in accordance with T9078-AS-GIB-010/271.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Strainers covered by this specification are intended for primarily for main steam systems, for protection of power equipment and throttle valves from dirt, sediment, pipe scale, and foreign objects. Strainers need to have service lives that are commensurate with the service life of a naval ship, and will be used in an environment that is subject to shock loadings associated with military service.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Composition required (see 1.2).

c. When either a single or dual outlet is required (see 3.2).

d. Design pressure and design temperature (see 3.2.1).

e. Maximum allowable temperatures (see 3.2.2).

f. Sizes of strainer assemblies, number of inlets and outlets, and shape (see 3.2.3).

g. End connections and flanges required (see 3.2.6).

h. Direction of steam flow (see 3.2.9).

i. AQL of 1.0 (see 4.1).

j. Packing required (see 5.1).

6.3 Specification effectivity. Current stock of strainers may be used until depleted; future requisitions for strainers should refer to the current version of this specification.

6.4 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.5 Subject term (key word) listing.

Connection, drain

Outlet, dual

Outlet, single

6.6 Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 4730-2019-086)

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